

Category	Item	Value	Unit
Material	Concrete	100	m³
	Steel	50	kg
	Brick	200	kg
	Wood	30	m³
Labor	Construction Worker	10	hr
	Electrician	5	hr
	Plumber	3	hr
	Painter	2	hr
Equipment	Excavator	1	hr
	Truck	2	hr
	Generator	1	hr
	Drill	1	hr

--	--	--	--	--

--	--	--	--

9702/34

October/November 2013

2 hours

Additional Materials: As listed in the Confidential Instructions.

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO NOT WRITE IN ANY BARCODES.

You are reminded of the need for good English and clear presentation in your answers.

You may lose marks if you do not show your working or if you do not use appropriate units.

Additional answer paper and graph paper should be used only if it becomes necessary to do so.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	
2	
Total	

This document consists of **12** printed pages.

You may not need to use all of the materials provided.

For
Examiner's
Use

- 1 In this experiment, you will investigate how the forces supporting a wooden strip change as the position of a suspended mass is moved.
- (a) (i) Suspend the single spring and the mass hanger from the rod of one of the clamps. Measure and record the height h_1 of the mass hanger above the bench, as shown in Fig. 1.1.

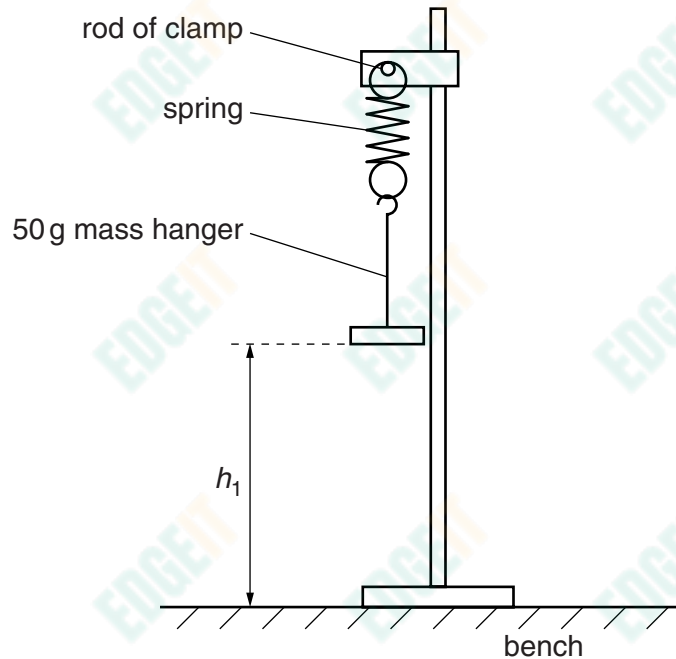


Fig. 1.1

$h_1 =$

- (ii) Add the 50 g slotted mass to the hanger. Measure and record the new height h_2 of the mass hanger above the bench.

$h_2 =$

[1]

- (iii) Calculate the change in height C , where $C = h_1 - h_2$.

$C =$

- (b) (i) Assemble the apparatus as shown in Fig. 1.2, with the mass M near to the middle of the wooden strip.

For
Examiner's
Use

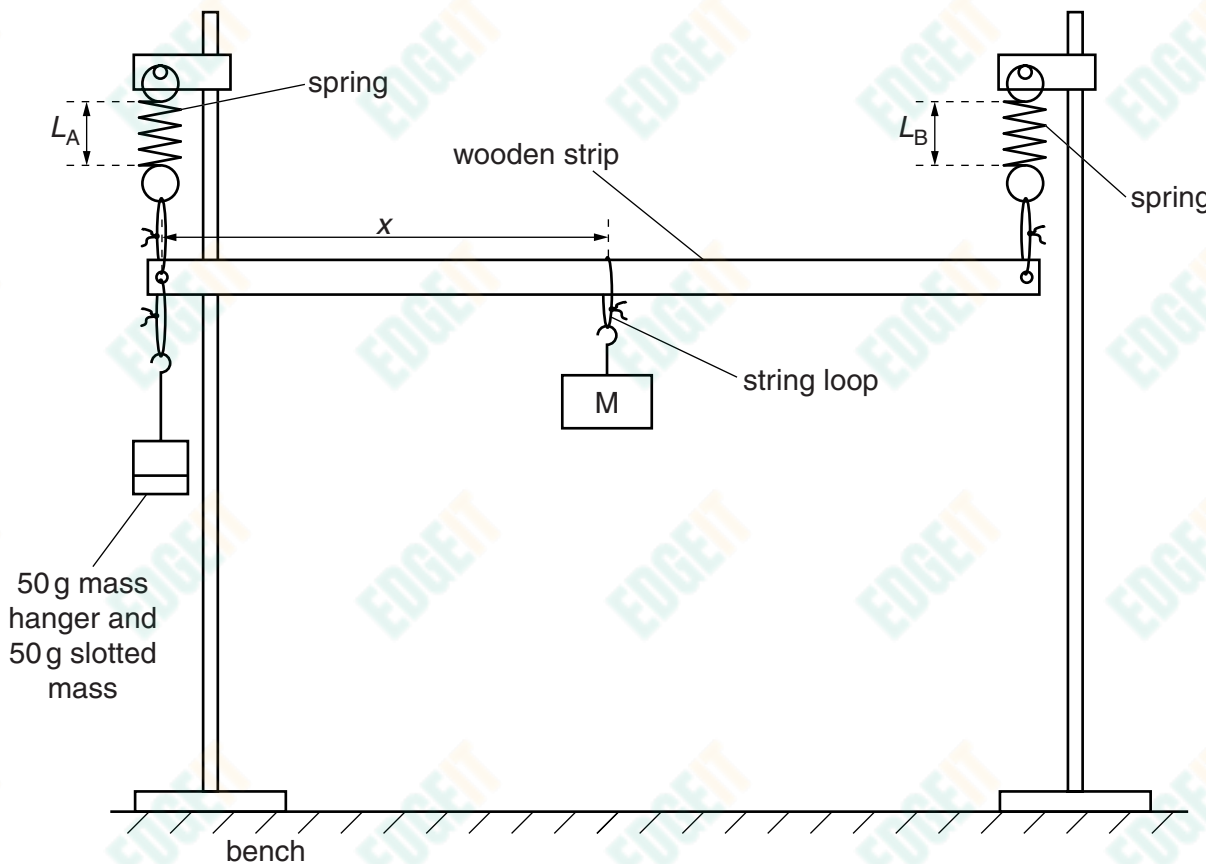


Fig. 1.2

- (ii) Adjust the apparatus so that the wooden strip is horizontal and the springs are vertical.
- (iii) Measure and record the distance x from the left-hand hole in the wooden strip to the string loop supporting M , as shown in Fig. 1.2.

$x =$

- (iv) Measure and record the length L_A of the coiled section of the left-hand spring, and the length L_B of the coiled section of the right-hand spring.

$L_A =$

$L_B =$

[1]

- (c) Reposition the string loop supporting M.
Repeat **(b)(ii)**, **(b)(iii)** and **(b)(iv)** until you have six sets of values of x , L_A and L_B .
Include values for $(L_A - L_B)$ and for $\frac{(L_A - L_B)}{C}$ in your table.

For
Examiner's
Use

☐

[10]

- (d) (i) Plot a graph of $\frac{(L_A - L_B)}{C}$ on the y -axis against x on the x -axis.

[3]

- (ii) Draw the straight line of best fit.

[1]

- (iii) Determine the gradient and y -intercept of this line.

gradient =

y -intercept =

[2]

For
Examiner's
Use

- (e) The quantities L_A , L_B , C and x are related by the equation

$$\frac{(L_A - L_B)}{C} = ax + b$$

where a and b are constants.

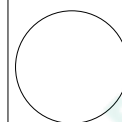
Use your answers from (d)(iii) to determine the values of a and b .
Give appropriate units.

$a = \dots\dots\dots$

$b = \dots\dots\dots$

[2]

For
Examiner's
Use



You may not need to use all of the materials provided.

For
Examiner's
Use

2 In this experiment, you will investigate the equilibrium of a spool of thread.

- (a) (i) Using the smaller of the two spools provided, take measurements to determine the outside diameter D and the inner diameter d , as shown in Fig. 2.1.

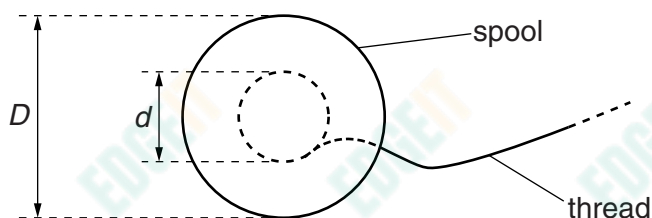


Fig. 2.1

$D =$ mm

$d =$ mm
[2]

- (ii) Calculate the ratio R , where $R = \frac{d}{D}$.

$R =$ [1]

- (b) Justify the number of significant figures that you have given for your value of R .

.....
.....
.....
.....[1]

- (c) Use the stand, boss and clamp to support the protractor as shown in Fig. 2.2.

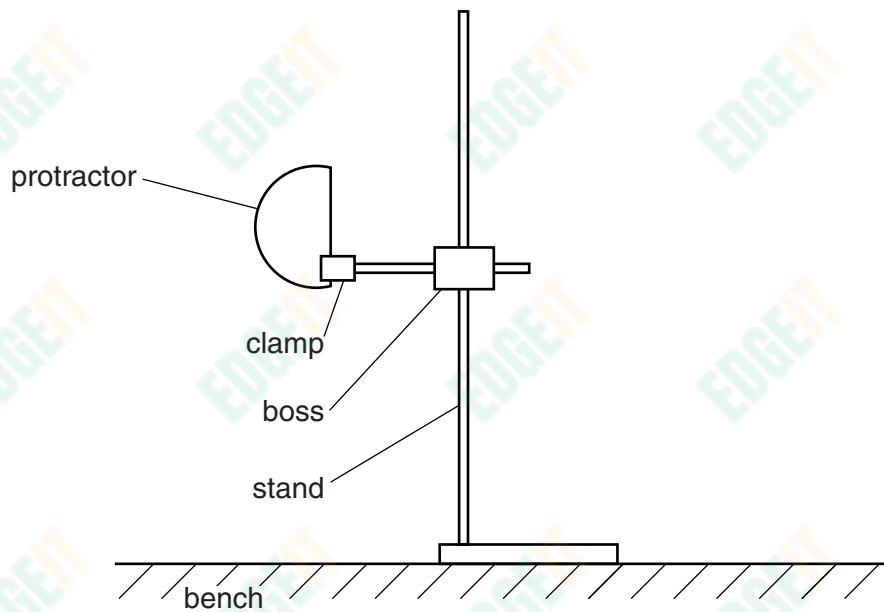


Fig. 2.2

- (d) (i) Using the smaller of the two spools, wrap the thread around its inner diameter until there is about 30cm of thread left unwound.
- (ii) With the spool on the bench, gently pull the thread vertically as shown in Fig. 2.3.

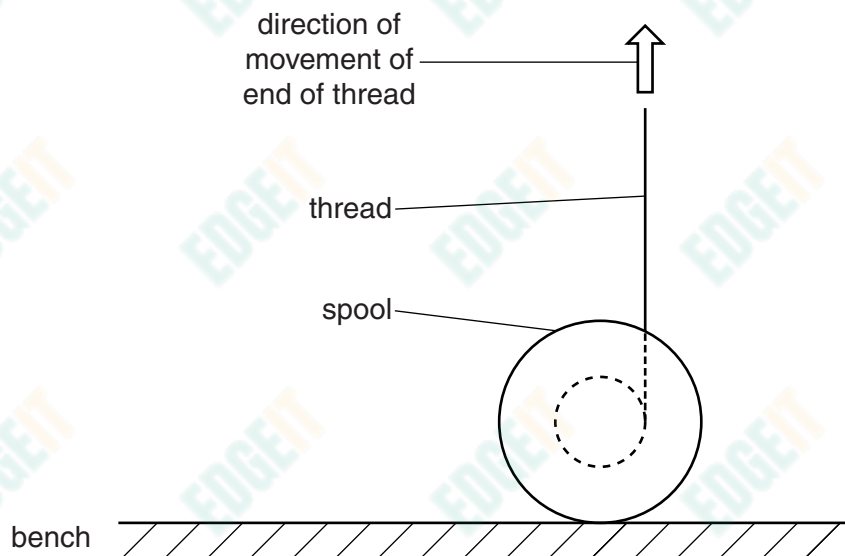


Fig. 2.3

- (iii) The spool will move to the left until the thread reaches a constant angle θ to the vertical with the spool rotating but not moving along the bench, as shown in Fig. 2.4.

For
Examiner's
Use

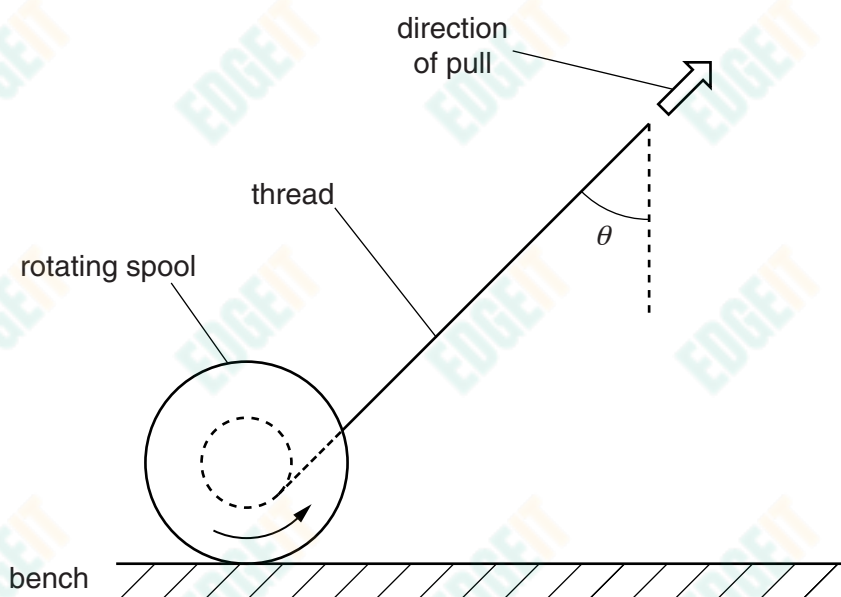


Fig. 2.4

- (iv) Measure and record the angle θ .

$\theta =$ [2]

- (e) Estimate the percentage uncertainty in your value of θ .

percentage uncertainty =[1]

- (f) Repeat (a) and (d) but using the larger of the two spools.

For
Examiner's
Use

$D =$

$d =$

$R =$

$\theta =$

[3]

- (g) It is suggested that the relationship between θ and R is

$$\sin \theta = kR$$

where k is a constant.

- (i) Using your data, calculate two values of k .

first value of $k = \dots\dots\dots$

second value of $k = \dots\dots\dots$

[1]

- (ii) Explain whether your results support the suggested relationship.

.....

 [1]

For
Examiner's
Use

- (h) (i) Describe four sources of uncertainty or limitations of the procedure for this experiment.

For
Examiner's
Use

1.
.....

2.
.....

3.
.....

4.
.....

[4]

- (ii) Describe four improvements that could be made to this experiment. You may suggest the use of other apparatus or different procedures.

1.
.....

2.
.....

3.
.....

4.
.....

[4]

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

University of Cambridge International Examinations is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of University of Cambridge Local Examinations Syndicate (UCLES), which is itself a department of the University of Cambridge.